Final Exam Review

Module 8 - Nonlinear Pharmacokinetics · Review set · 19 Oct 2026

Name _____ Date _____

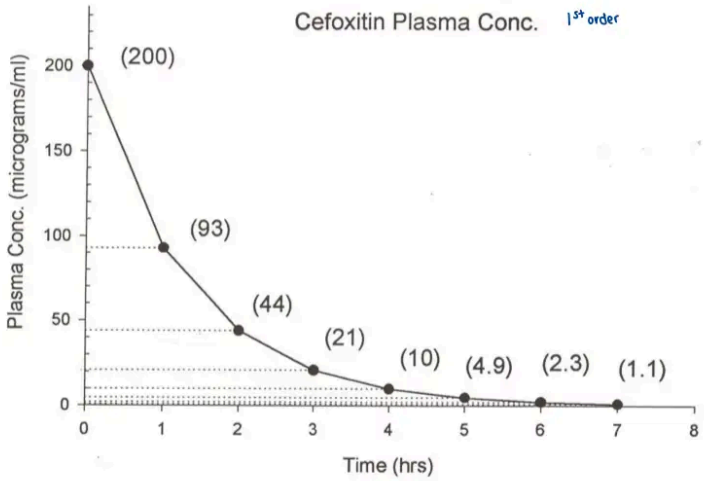
Slide 3 Which of the following is characteristic of first-order processes?

A. units of k of (concentration or amount)/time
B. a constant rate of elimination
C. dC/dt = -k
D. a constant half-life

Slide 7 400 mg of a drug was administered by IV bolus injection. The equation describing the drug in the body is given as $C_p = (40\ mg/L)e^{-0.9t}$. What is the apparent volume of distribution of this drug?

A. 10 L
B. 40 L
C. 20 L
D. 50 L

Slide 8 Cefoxitin plasma concentrations were 200, 93, 44, 21, 10, 4.9, 2.3 and 1.1 micrograms/mL at 0, 1, 2, 3, 4, 5, 6 and 7 hours, as shown. Estimate the half-life of cefoxitin.



20

Slide 10 The renal clearance of an antibiotic was estimated to be 45 mL/min. What can be inferred from this information?

A. The drug is filtered and reabsorbed.

B. The drug is filtered and actively secreted.

C. There is some evidence of renal impairment.

Slide 11 Hepatic clearance is most readily approximated with creatinine clearance.

A. False

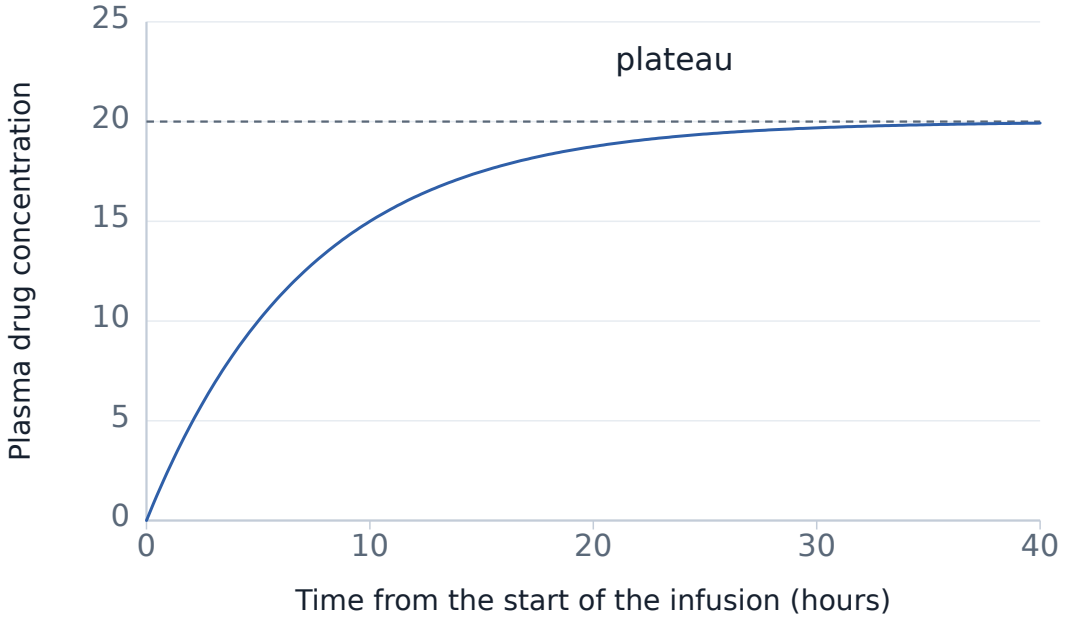
B. True

Slide 13 What is the expected maximum steady state plasma concentration following multiple IV bolus dosing every 12 hours of 300 mg of a drug with a t½ of 4 hours and V_D of 16 L?

A. None of the above
B. 32.6 mg/L
C. 21.4 mg/L
D. 18.7 mg/L

Slide 13 A drug with a t½ of 4 hours and V_D of 16 L is given as multiple IV bolus doses of 300 mg every 12 hours. What is the minimum plasma concentration at steady state?

Slide 15 Which type of drug input is depicted by this curve?



A. instantaneous input
B. zero-order input
C. first-order input

Slide 15 Match each pattern of drug input and output to the type of administration that gives it.

1. Instantaneous input, first-order output _____
2. First-order input, first-order output _____
3. Zero-order input, first-order output _____

A. IV bolus
B. Conventional oral dose
C. IV infusion

Slide 17 What is the bioavailability of a 500 mg tablet which yielded an AUC of 115 (mg/L)hr compared to a 400 mg IV bolus dose that yielded an AUC of 132 (mg/L)hr?

A. 50%
B. 92%
C. 70%
D. 85%

Slide 17 A 500 mg tablet yielded an AUC of 115 (mg/L)hr and a 400 mg IV bolus dose yielded an AUC of 132 (mg/L)hr. What oral dose would provide similar therapeutic activity as the 400 mg bolus dose? Round to the nearest 5.

Slide 19 Drugs that display Michaelis-Menten saturation kinetics will often exhibit zero-order kinetics at low concentrations and first-order kinetics at high concentrations.

A. False
B. True

Slide 21 The trapezoidal rule provides a simple way to estimate the AUC.

A. False
B. True

Slide 23 What is the duration of action for a 1 g IV bolus dose with a V_D = 12.5 L, t½ = 4hr and MEC= 20 mg/L ?

A. None of the above
B. 4 hours
C. 12 hours
D. 8 hours

Slide 23 A drug has a V_D = 12.5 L, t½ = 4hr and MEC= 20 mg/L. If 500 mg were given as an IV bolus dose instead of 1 g, what would the duration of action be?

Slide 25 What is the plasma concentration at the end of a 2 hour infusion of a drug with a t½ of 3 hours and a V_D of 18 L at a rate of 50 mg/hr?

A. 3.03 mg/L
B. 4.45 mg/L
C. 4.81 mg/L
D. 31.5 mg/L

Slide 25 A drug with a t½ of 3 hours and a V_D of 18 L is infused at a rate of 50 mg/hr for 2 hours. What is the concentration 3 hours after cessation of the infusion?

Slides 25 and 26

A drug with a t½ of 3 hours and a V_D of 18 L is infused at a rate of 50 mg/hr for 2 hours. Another 2 hour infusion was started 6 hours after the start of the first infusion. What is the concentration 4 hours after cessation of the second infusion?

Slide 25 A drug with a t½ of 3 hours and a V_D of 18 L is infused at a rate of 50 mg/hr. If the infusion were continued at this rate, what would the steady-state concentration be?

Slide 25 A drug with a t½ of 3 hours and a V_D of 18 L is infused. If it were infused at a rate of 100 mg/hr, what would the steady-state concentration be?

Slide 25 A drug with a t½ of 3 hours infused at 50 mg/hr reaches a steady-state concentration of about 12 mg/L. Infused at 100 mg/hr, it reaches about 24 mg/L. How long does it take to reach 24 mg/L at 100 mg/hr, compared with reaching 12 mg/L at 50 mg/hr?

A. Twice the time
B. Half the time
C. The same time